Work, Force, Energy & Simple Machines (Part - 1)

Work Force and Energy

FORCE

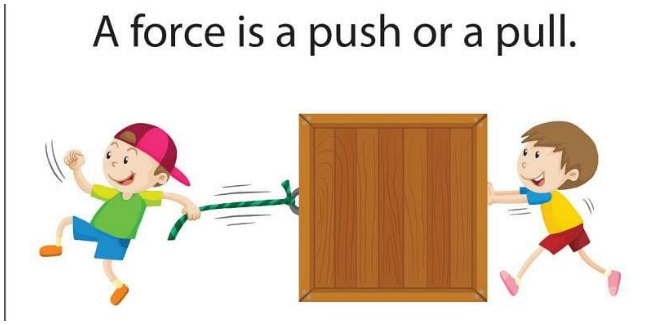


Fig: *Force*

A force is a push or pull. It can help us to do the work. It brings the change or tries to bring the change in the state of motion. Force may be of various types, e.g. mechanical, gravitation, magnetic, frictional, electric and buoyant force.

WORK

Whenever an applied force is capable to move an object on which the force is applied, the work is done. No change in position means no work. Work is measured in joule, which means the object is displaced by one metre with one Newton Force.

SIMPLE MACHINE

It is a tool that makes the work easy. For example

- Lever
- Pulley
- Wheel and axle
- Inclined plane
- Screw
- Wedge

LEVER

A lever is a simple machine. It is a rigid rod that turns around a fixed point called fulcrum. It consists of fulcrum (f), load (w) and effort.

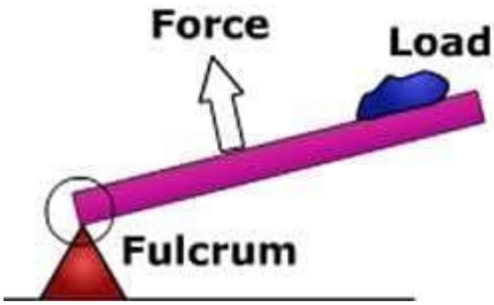
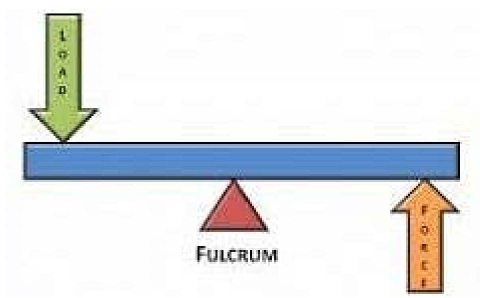
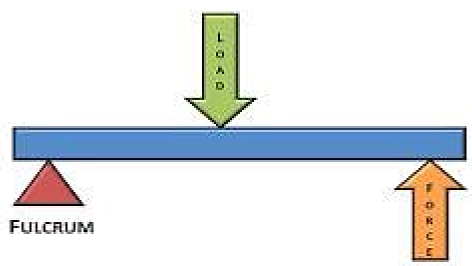
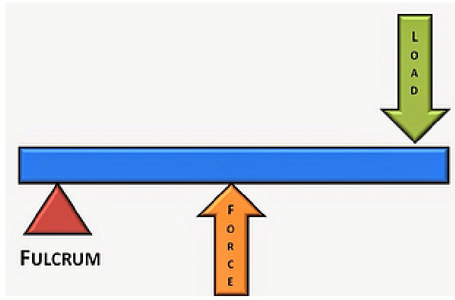


Fig: A Lever

Levers classified into three kinds according to the position of fulcrum, load and effort.

Type of Lever

1st Class Fulcrum between load and effort		E.g. Scissors Claw-hammer
2nd Class Load between fulcrum and effort		E.g. Nutcraker Wheel barrow

3rd Class Effort between load and fulcrum	 	E.g. Finishing rod Broom

PULLEY

The pulley is a wheel with a grooved around its rim. The wheel turns around an axle. A belt or string fits into groove. There are two types of pulleys, a fixed pulley and a movable pulley. The pulley used for drawing water from a well is fixed pulley, movable pulleys along with fixed pulleys are used to lift woods.

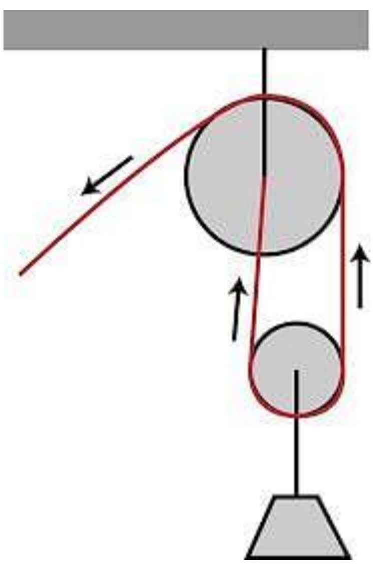


Fig: *A Pulley*

WHEEL AND AXEL

- The wheel and axle consists of a wheel attached to a smaller axle so that these two parts rotate together in which a force is transferred from one to the other.

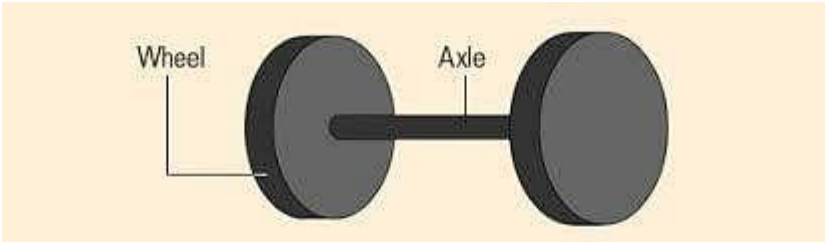


Fig: A picture showing a wheel attached to an axle.

INCLINED PLANE

- An inclined plane, also known as a ramp, is a flat supporting surface tilted at an angle, with one end higher than the other, used as an aid for raising or lowering a load.

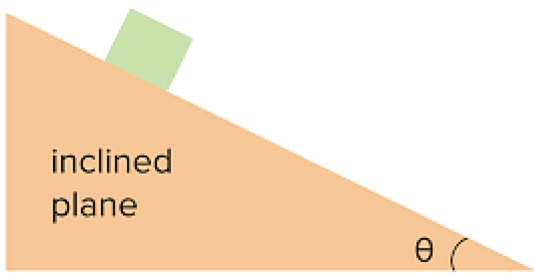


Fig: An Inclined Plane

SCREW

A short, slender, sharp-pointed metal pin with a raised helical thread running around it and a slotted head, used to join things together by being rotated so that it pierces wood or other material and is held tightly in place.

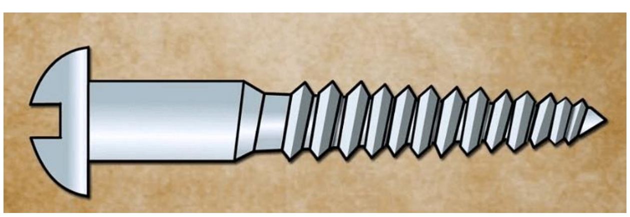


Fig: A Screw

WEDGE

A wedge consists of two inclined planes that meet at a sharp edge.

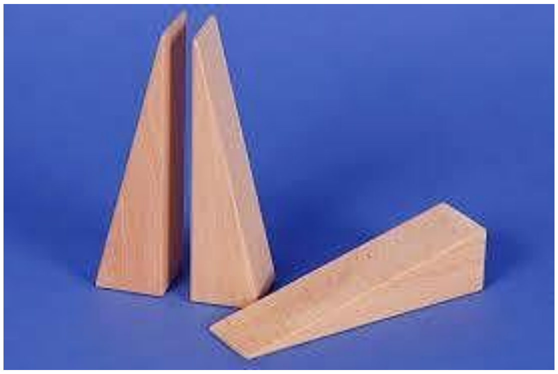


Fig: A Wedge with a sharp edge.

ENERGY

- Energy is the capacity to do the work. Energy is neither created not destroyed but only transformed in another form. Total energy is always fixed.
- Renewable source of energy, which are constantly replenished by nature, e.g. Sun, wind and water.
- **Non-renewable source of energy:** Takes millions of years to be formed. These sources of energy are called non- renewable source of energy, e.g. Fossil fuels (Petrol, diesel) they are limited in amount and will be exhausted one day.
- Work

Introduction:
When a force is applied on a body, it may or may not move. If the applied force is sufficient to move the object from one place to another, it is said that the work has been done. If the applied force is not sufficient to move the object from one place to another, then no work has been done. Energy is the capacity to do work. In this chapter we will study about the work and energy.

Work
Whenever an applied force is capable to move an object, on which the force is applied, the work is done. A boy moves a chair from one place to another by applying force that means the work is done. The same boy applies a force on the wall to move it from one place to another and wall cannot move, hence, no work is done.

The following are the conditions for the work to be done:

1. A force should be applied on a body.

The body on which the force is applied should change its position.

Image

In the picture above a body "A", moves from one place to another by the application of force. Therefore, the work is done.

Image

Unit for the Measurement of Work

The unit for the measurement of work is joule which is also expressed by 'J'. The unit of work is derived from the product of applied force on a body and displacement of the body due to the applied force. Therefore, we can say that if a force of 1 N acting on a body and body moves 1 m in the direction of the applied force then 1 J of work is done. Hence, according to the work done, 1 J = 1 N (Applied force) x 1 m (Displacement) = Mm (Newton metre).

Commonly Asked Questions:

1. Find the work done if a body moves through a distance of 5 m in the direction of applied force of 100 N.
(a) 500J
(b) 400J
(c) 1000J
(d) 300J
(e) None of these
Answer: (a)

Explanation
Work done = Force x Displacement = 100 N x 5 m = 500 Nm = 500 J.
Therefore, option A is correct and rest of the options is incorrect.

How much work is done on lifting a body of 1 kg up to the distance of 1 m above the ground or vertically?
(a) 5.6 J
(b) 2.3 J
(c) 100 J
(d) 9.8 j
(e) None of these
Answer: (d)

Explanation
When a body of 1 kg wt. is lifted upward, it require the force of 9.8 N.
Hence, the work done = Force x Displacement = 9.8 N x l m = 9.8 Nm = 9.8 J.
Therefore, option D is correct and rest of the options is incorrect